UTS Sistem Basis Data

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6.10 Disini kita akan membuat table Hotel dengan integrity enhancement features

```
uts_basis_data=# create extension if not exists btree_gist;
CREATE EXTENSION
```

```
uts_basis_data=# create table hotel (
uts_basis_data(# hotelNo varchar(15) primary key,
uts_basis_data(# hotelName varchar(40) not null,
uts_basis_data(# city varchar(40) not null
uts_basis_data(# );
```

6.11 Membuat table Room, Booking, Guest dengan catatan

(a) type must be one of Single, Double, or Family.
(b) price must be between £10 and £100.
(c) roomNo must be between 1 and 100.
(d) dateFrom and dateTo must be greater than today's date.

(e) The same room cannot be double-booked.
(f) The same guest cannot have overlapping bookings.

```
uts_basis_data=# create table room (
uts_basis_data(# roomNo int not null,
uts_basis_data(# hotelNo varchar(15) not null,
uts_basis_data(# type varchar(15) not null,
uts_basis_data(# price decimal(5,2) not null,
uts_basis_data(# primary key (roomNo, hotelNo),
uts_basis_data(# foreign key (hotelNo) references hotel(hotelNo) on delete cascade,
uts_basis_data(# constraint chk_room_type check (type in ('Single', 'Double', 'Family')),
uts_basis_data(# constraint chk_room_price check (price between 10 and 100),
uts_basis_data(# constraint chk_room_no check (roomNo between 1 and 100)
uts_basis_data(# );
CREATE TABLE
```

disini kita membuat table room dengan syarat room tidak boleh dibooking 2 kali dalam tanggal yang sama

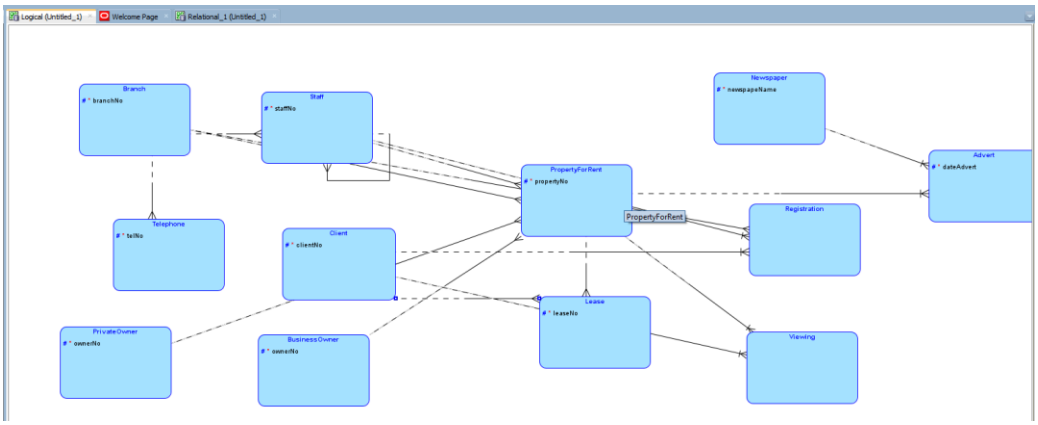
```
uts_basis_data=# create table guest (
uts_basis_data(# guestNo varchar(15) primary key,
uts_basis_data(# guestName varchar(40) not null,
uts_basis_data(# guestAddress varchar(110) not null
uts_basis_data(# );
CREATE TABLE
```

```
uts_basis_data=#  create table booking (
uts_basis_data(# hotelNo varchar(15) not null,
uts_basis_data(# guestNo varchar(15) not null,
uts_basis_data(# dateFrom date not null,
uts_basis_data(# dateTo date not null,
uts_basis_data(# roomNo int not null,
uts_basis_data(# primary key (hotelNo, guestNo, dateFrom),
uts_basis_data(# foreign key (hotelNo) references hotel(hotelNo) on delete cascade,
uts_basis_data(# foreign key (guestNo) references guest(guestNo) on delete cascade,
uts_basis_data(# foreign key (roomNo, hotelNo) references room(roomNo, hotelNo) on delete cascade,
uts_basis_data(# constraint chk_dates_future check (dateFrom > current_date and dateTo > current_date),
uts_basis_data(# constraint chk_date_logic check (dateTo > dateFrom),
uts_basis_data(# constraint exclude_room_overlap exclude using gist (
uts_basis_data(# roomNo with =,
uts_basis_data(# hotelNo with =,
uts_basis_data(# daterange(dateFrom, dateTo) with &&
uts_basis_data(# ),
uts_basis_data(# constraint exclude_guest_overlap exclude using gist (
uts_basis_data(# guestNo with =,
uts_basis_data(# daterange(dateFrom, dateTo) with &&
uts_basis_data(# )
uts_basis_data(# );
```

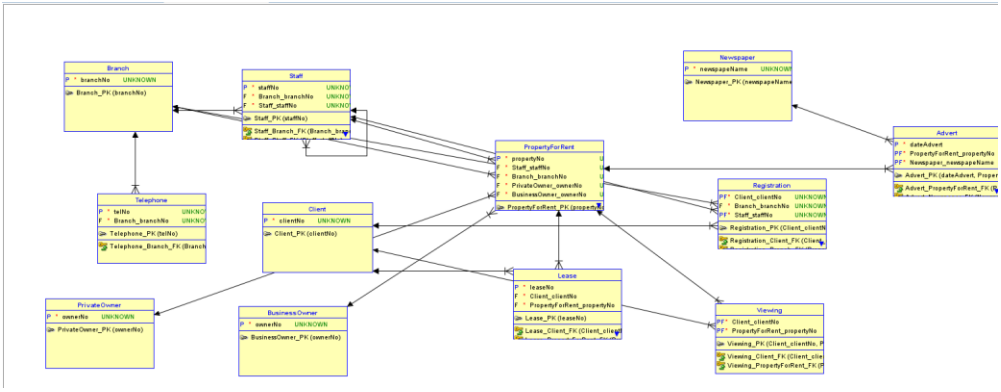
Disini kita membuat table booking yang dimana guest yang sama tidak bisa booking 2 tempat berbeda, dan bisa menempatkan tanggal agar jadwal yang sudah dibook tidak bisa dibook lagi

16.1 Membuat Logical Model kemudian diubah menjadi physical model dan generate DDL kemudian DUMP di postgres

Logical Model



Physical Model



DDL dalam query Oracle 11g belum diubah ke query Postgres

```
CREATE TABLE Advert
    (
     dateAdvert                 UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     PropertyForRent_propertyNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Newspaper_newspapeName     UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Advert
    ADD CONSTRAINT Advert_PK PRIMARY KEY ( dateAdvert, PropertyForRent_propertyNo, Newspaper_newspapeName ) ;

CREATE TABLE Branch
    (
     branchNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Branch
    ADD CONSTRAINT Branch_PK PRIMARY KEY ( branchNo ) ;

CREATE TABLE BusinessOwner
    (
     ownerNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE BusinessOwner
    ADD CONSTRAINT BusinessOwner_PK PRIMARY KEY ( ownerNo ) ;

CREATE TABLE Client
```

```
CREATE TABLE Client
    (
     clientNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
;

ALTER TABLE Client
    ADD CONSTRAINT Client_PK PRIMARY KEY ( clientNo ) ;

CREATE TABLE Lease
    (
     leaseNo                    UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     Client_clientNo            UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     PropertyForRent_propertyNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
;

ALTER TABLE Lease
    ADD CONSTRAINT Lease_PK PRIMARY KEY ( leaseNo ) ;

CREATE TABLE Newspaper
    (
     newspapeName UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
;

ALTER TABLE Newspaper
    ADD CONSTRAINT Newspaper_PK PRIMARY KEY ( newspapeName ) ;

CREATE TABLE PrivateOwner
    (
     ownerNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
;

ALTER TABLE PrivateOwner
    ADD CONSTRAINT PrivateOwner_PK PRIMARY KEY ( ownerNo ) ;

CREATE TABLE PropertyForRent
```

```
CREATE TABLE PropertyForRent
    (
     propertyNo            UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     Staff_staffNo         UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     Branch_branchNo       UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     PrivateOwner_ownerNo  UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     BusinessOwner_ownerNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
;

ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_PK PRIMARY KEY ( propertyNo ) ;

CREATE TABLE Registration
    (
     Client_clientNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     Branch_branchNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     Staff_staffNo   UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
;

ALTER TABLE Registration
    ADD CONSTRAINT Registration_PK PRIMARY KEY ( Client_clientNo, Staff_staffNo ) ;

CREATE TABLE Staff
    (
     staffNo         UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     Branch_branchNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     Staff_staffNo   UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
```

```
ALTER TABLE Staff
    ADD CONSTRAINT Staff_PK PRIMARY KEY ( staffNo ) ;

CREATE TABLE Telephone
    (
     telNo           UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     Branch_branchNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
;

ALTER TABLE Telephone
    ADD CONSTRAINT Telephone_PK PRIMARY KEY ( telNo ) ;

CREATE TABLE Viewing
    (
     Client_clientNo            UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL ,
     PropertyForRent_propertyNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                     NOT NULL
    )
;

ALTER TABLE Viewing
    ADD CONSTRAINT Viewing_PK PRIMARY KEY ( Client_clientNo, PropertyForRent_propertyNo ) ;

ALTER TABLE Advert
    ADD CONSTRAINT Advert_Newspaper_FK FOREIGN KEY
    (
     Newspaper_newspapeName
    )
    REFERENCES Newspaper
    (
     newspapeName
    )
;

ALTER TABLE Advert
    ADD CONSTRAINT Advert_PropertyForRent_FK FOREIGN KEY
    (
     PropertyForRent_propertyNo
    )
    REFERENCES PropertyForRent
    (
     propertyNo
```

```
ALTER TABLE Lease
    ADD CONSTRAINT Lease_Client_FK FOREIGN KEY
    (
     Client_clientNo
    )
    REFERENCES Client
    (
     clientNo
    )
;

ALTER TABLE Lease
    ADD CONSTRAINT Lease_PropertyForRent_FK FOREIGN KEY
    (
     PropertyForRent_propertyNo
    )
    REFERENCES PropertyForRent
    (
     propertyNo
    )
;

ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_Branch_FK FOREIGN KEY
    (
     Branch_branchNo
    )
    REFERENCES Branch
    (
     branchNo
    )
;

--  ERROR: FK name length exceeds maximum allowed length(30)
ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_BusinessOwner_FK FOREIGN KEY
    (
     BusinessOwner_ownerNo
    )
    REFERENCES BusinessOwner
    (
     ownerNo
    )
;

--  ERROR: FK name length exceeds maximum allowed length(30)
ALTER TABLE PropertyForRent
```

```
-- ERROR: FK name length exceeds maximum allowed length(30)
ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_PrivateOwner_FK FOREIGN KEY
    (
     PrivateOwner_ownerNo
    )
    REFERENCES PrivateOwner
    (
     ownerNo
    )
;

ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_Staff_FK FOREIGN KEY
    (
     Staff_staffNo
    )
    REFERENCES Staff
    (
     staffNo
    )
;

ALTER TABLE Registration
    ADD CONSTRAINT Registration_Branch_FK FOREIGN KEY
    (
     Branch_branchNo
    )
    REFERENCES Branch
    (
     branchNo
    )
;

ALTER TABLE Registration
    ADD CONSTRAINT Registration_Client_FK FOREIGN KEY
    (
     Client_clientNo
    )
    REFERENCES Client
    (
     clientNo
    )
;

ALTER TABLE Registration
    ADD CONSTRAINT Registration_Staff_FK FOREIGN KEY
    (
     Staff_staffNo
    )
    REFERENCES Staff
    (
```

```
ALTER TABLE Staff
    ADD CONSTRAINT Staff_Branch_FK FOREIGN KEY
    (
     Branch_branchNo
    )
    REFERENCES Branch
    (
     branchNo
    )
;

ALTER TABLE Staff
    ADD CONSTRAINT Staff_Staff_FK FOREIGN KEY
    (
     Staff_staffNo
    )
    REFERENCES Staff
    (
     staffNo
    )
;

ALTER TABLE Telephone
    ADD CONSTRAINT Telephone_Branch_FK FOREIGN KEY
    (
     Branch_branchNo
    )
    REFERENCES Branch
    (
     branchNo
    )
;

ALTER TABLE Viewing
    ADD CONSTRAINT Viewing_Client_FK FOREIGN KEY
    (
     Client_clientNo
    )
    REFERENCES Client
    (
     clientNo
    )
;

ALTER TABLE Viewing
    ADD CONSTRAINT Viewing_PropertyForRent_FK FOREIGN KEY
    (
     PropertyForRent_propertyNo
    )
    REFERENCES PropertyForRent
```

```
ALTER TABLE Viewing
    ADD CONSTRAINT Viewing_PropertyForRent_FK FOREIGN KEY
    (
     PropertyForRent_propertyNo
    )
    REFERENCES PropertyForRent
    (
     propertyNo
    )
;
```

Kemudian kita akan DUMP ke postgres dengan mengubah sedikit query untuk mencocokkan ke query postgres